Data Analysis of Metal Genres

*This project is made for INF356

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***Abstract*—Metal music is often perceived as a homogeneous genre despite its rich diversity of styles and subgenres. This study investigates whether metal genres are acoustically distinguishable in a statistically meaningful way by analyzing a manually curated dataset of 2,673 tracks representing 21 main genres and numerous subgenres. The dataset combines Spotify popularity scores with detailed audio features obtained via the ReccoBeats API, enabling a comprehensive analysis of musical characteristics that are no longer directly accessible through Spotify. Descriptive statistical analysis is first employed to summarize central tendencies and distributions of key audio features across genres. Inferential methods, including one-way ANOVA and post-hoc pairwise tests, are then applied to assess whether observed differences between genres are statistically significant. Additionally, a chi-square test is used to examine the relationship between musical mode (major/minor) and genre membership. Finally, a multiple linear regression model is fitted to explore the extent to which acoustic features can predict track popularity. The results demonstrate that most audio features vary significantly across metal genres, providing strong evidence that metal music is acoustically diverse rather than uniform. Regression findings further indicate that popularity is partially associated with specific acoustic attributes, although a substantial portion of variance remains unexplained. Overall, this study highlights the value of combining descriptive and inferential data analysis techniques to quantitatively examine musical genre distinctions.**

I. Introduction

In this document, we will describe and analyze our data set to compare the acoustic features of different genres and subgenres of metal music.

Overview

Metal music is one of the most distinctive and popular genres in the music world. Throughout history, it has evolved into many different genres and subgenres, each quite different from the others.

In this context, the project will first provide a brief history of metal music and the emergence of different genres, followed by a comparison and analysis of the acoustic characteristics of these genres.

For this purpose, a manually created dataset has been prepared by extracting data from Spotify's database on the most popular and genre-defining songs of the most important representative bands of each metal genre. This data set will be examined using various data analysis methods, and conclusions will be drawn from the results.

A. Short History of Heavy Metal:

Heavy metal (or simply metal) is a genre of rock music that developed in the late 1960s and early 1970s, largely in the United Kingdom and United States. It has its roots in blues, rock, and 1960s psychedelic rock.

The late 1960s was a period when rock music became increasingly aggressive, sound levels rose, and guitar distortion began to become standard. During these years, bands like Led Zeppelin, Deep Purple, and Jimi Hendrix Experience laid the foundation for "hard rock" by transforming blues-based rock into a more aggressive, louder, and technically more intense form. Some also refer to this period as "proto-metal."

However, during this period, the terms "hard rock" and "heavy metal" were often used interchangeably; a clear distinction between genres had not yet been established. Most music critics and listeners defined anything with a heavy, distorted guitar sound as "heavy metal," but this usage remained quite broad and vague. Heavy metal truly differentiated itself in the early 1970s with Black Sabbath's dark, slow, and ominous riff structure.Today, Black Sabbath is widely recognized by many music critics as the first "heavy metal" band.

The gloomy atmosphere, thematic depth, and unique rhythmic structure characteristic of Sabbath's style formed the first aesthetic framework that clearly distinguished heavy metal from hard rock. Thus, the hardening rock movement of the late 1960s split into two main paths by the time the 1970s arrived: hard rock, which retained its blues foundations and more melodic structure, and heavy metal, built on dark tones, heavy riffs, and a unique dramatic aesthetic.

The term "heavy metal" is used in two different ways: to describe early metal bands, or as a general term encompassing all other metal genres.

B. Emergence of Different Genres :

After heavy metal established its own distinct identity in the 1970s, the genre's development rapidly diversified through a combination of musical experimentation, regional scenes, and cultural influences.

By the late 1970s, the "New Wave of British Heavy Metal" (NWOBHM) movement in the UK, featuring bands like Iron Maiden, Judas Priest and Saxon, accelerated the tempo of

metal and shifted it towards a more melodic and technical structure.

During the same period, the thrash metal scene emerged in the United States; bands like Metallica, Slayer, Megadeth, and Anthrax -these bands also referred to as "The Big Four"- added radical intensity to metal music with their speed, aggressive riffs, and punk influences.

Throughout the 1980s, extreme subgenres emerged one after another: Death metal developed with brutal vocals and complex riffs on the Florida scene, while the black metal wave in Norway created one of the most distinctive sounds in metal history with its lo-fi production, dark atmosphere, and radical themes such as satanism and anti-christianity.

From the 1990s onward, the proliferation of subgenres gained even more momentum: doom metal with its slow and heavy structure; gothic metal with its dark romanticism and atmosphere; and power metal with its epic melodies and high vocals each established their own esthetic identities.

During the same period, hybrid genres such as alternative metal, nu metal, and metalcore emerged from the fusion of metal with rap, electronic, and hardcore punk, reaching a wide audience. Today, metal music is divided into hundreds of subgenres that span a wide range in terms of technique, theme, and production, and continues to evolve.

Motivation

As mentioned above, metal music has evolved in many different ways throughout history. Although the average listener may consider any music with electric guitar sounds and distortion to be metal, metal is actually much more than that. Randomly selected songs from two different metal genres will sound very different from each other.

In this context, this project aims to demonstrate that each metal music genre is acoustically distinct from the others. This will help us show how inaccurate the comment "they all sound the same" made by some uninformed circles about metal music really is. Additionally, by comparing the various subgenres of metal music, we will explore how each genre differs in specific aspects, what distinct characteristics each genre possesses, and how these genres relate to other genres.

Research Questions

In this section we will try to divide our problem to smaller parts and try to gather info for answering the main question.

Main Question: Are metal music genres acoustically different from each other in a meaningful way?

We want to find out whether there are common acoustic characteristics that define songs of a certain metal genre, and if so, whether these characteristics are specific to that genre or belong to metal music in general.

Sub-questions:

- What is the distribution of the fundamental acoustic characteristics (tempo, energy, valence, pitch, danceability, etc.) of each metal subgenre? : We want to know whether all metal music genres generally share certain acoustic characteristics or whether each genre is distinct enough to be defined as unique. To do this, we will analyze the acoustic features of songs of each genre and examine whether there are commonalities among them.
- Are certain acoustic characteristics statistically significantly higher or lower within a specific subgenre and if so what is the reason behind that ? : Some acoustic characteristics may only differ in certain subgenres; this may be due to the origin of that genre, as some metal music genres have been greatly influenced by other non-metal music genres.
- Are the acoustic characteristics evenly distributed within the genre, or can there be significant differences from song to song even within a particular genre ? : Here, we will look at the distribution of the acoustic characteristics that distinguish certain genres from other genres within the genre itself. Songs of that genre may have similar characteristics, or they may be randomly distributed.
- Are subgenres that appear similar (e.g., Death Metal vs. Black Metal, Sludge vs. Doom) really close to each other acoustically ? : Some genres may be thought to be very similar to each other, both in terms of their historical origins and the time they were first heard by listeners. We want to find out whether genres that are thought to be similar have similar acoustic characteristics.
- Is there a relationship between popularity (Spotify popularity score) and acoustic characteristics ? : The dataset also includes a popularity score assigned by Spotify for each song. We want to know if there are common acoustic characteristics that determine the popularity of a genre.

Contributions

This project makes several methodological and empirical contributions within the scope of introductory data analysis applied to music data.

First, the study introduces a manually curated, balanced dataset representing the metal music ecosystem, consisting of 2,673 tracks labeled with both main genre and subgenre information. Unlike automatically generated datasets, this dataset was carefully constructed to reflect genre-defining artists and stylistic diversity within metal music, ensuring higher reliability and interpretability for statistical analysis.

Second, the project contributes an analysis-ready dataset by integrating Spotify-derived popularity scores with detailed audio features obtained from the ReccoBeats API into a single unified table. This integration enables simultaneous examination of musical structure and listener reception, which is rarely available in a clean and consistent format for metal music.

Third, the project provides a comprehensive descriptive statistical analysis of metal music across a wide range of acoustic features. Measures such as mean, median, standard deviation, and distributional visualizations (histograms and boxplots) were systematically applied, offering a clear quantitative characterization of how metal genres differ in

terms of energy, tempo, loudness, danceability, acousticness, instrumentalness, speechiness, valence, and mode.

Fourth, beyond descriptive analysis, the study applies formal statistical inference techniques to test whether observed differences are statistically meaningful. One-way ANOVA was used to evaluate genre-level differences across multiple audio features, followed by post-hoc pairwise comparisons to identify which genres differ significantly. Additionally, a chi-square test was employed to examine the relationship between musical mode (major/minor) and genre, extending the analysis to categorical data.

Fifth, the project explores the relationship between acoustic features and popularity through multiple linear regression. By modeling popularity as a function of audio characteristics and genre indicators, the analysis demonstrates how statistical modeling can be used to quantify associations, evaluate effect directions, and assess predictive performance using standard regression metrics.

Finally, the results collectively contribute to a data-driven evaluation of common claims about metal music, showing that metal genres are not only descriptively different but also statistically distinguishable based on measurable audio properties. The project illustrates how foundational data analysis concepts taught in the course—descriptive statistics, hypothesis testing, ANOVA, and regression—can be coherently combined to answer real-world research questions.

Overall, this work serves as a structured example of applying introductory data analysis methodologies to a complex, real-world dataset, bridging theoretical concepts with practical interpretation.

II. Method and Our Data

This section explains why this dataset was selected, describes its characteristics, content, attributes, and summarizes the methods that will be used to answer the research questions.

A. Story / Overview of Our Data

Metal music is one of the most diverse musical ecosystems, containing hundreds of subgenres that differ dramatically in sound, structure, intensity, production style, history and cultural background. Because of this complexity, publicly available datasets are often too limited, unbalanced, or inconsistent to allow meaningful comparison between genres.

At the same time, to which genre a song belongs and how many genres and subgenres there are is also a highly debatable and open to interpretation topic. Although we initially considered selecting songs based on Spotify's own genre tags, we decided against it for the following reasons:

- Many metal subgenres share overlapping tags.
- Niche genres return incomplete or incorrect data.
- Spotify often limits the number of returned tracks (e.g., max. 20 per search).
- Some tags merge unrelated bands under the same "genre" label.

To overcome these limitations, a custom, manually curated dataset was constructed.

First, a list of metal genres and subgenres was created based on authoritative sources such as Wikipedia's Metal Genre Tree, Encyclopaedia Metallum, and various metal music communities (Reddit, RateYourMusic, etc.).

For each subgenre, its most influential and well-known representative artists were identified manually. Rather than selecting random tracks, we chose the top 10 most popular tracks of each artist through Spotify's API. This ensures that our dataset reflects:

- Each genre's most characteristic sound,
- High-quality production representative of the style,
- A reliable sample of tracks that shaped the genre's identity.

The audio features (danceability, energy, loudness, valence, etc.) were extracted using ReccoBeats API -which appears to returns old spotify data-, since Spotify API no longer provides - unfortunately audio feature support has ended as of May 2025- full audio analysis of songs.

As a result, the final dataset contains 2673 tracks representing dozens of metal genres and subgenres, each labeled with following attributes:

- main genre
- subgenre
- artist name
- track name
- album name
- spotify track id
- popularity
- id
- href
- full audio feature vector which consists of several attributes

This dataset is rich, balanced, manually controlled, and designed specifically for analyzing acoustic differences across metal genres. The meanings of these attributes will be further explained in the next section.

B. Attributes of Our Data

The dataset has the following columns:

- main genre : The broader genre category (e.g., Death Metal, Black Metal, Doom Metal).
- subgenre : More specific stylistic classification (e.g., Symphonic Black Metal, Melodic Death Metal).
- artist name : The exact Spotify name of the artist.
- track name : The name of the song.
- album name : The album from which the track originates.
- spotify track id : A unique ID used to retrieve detailed information from APIs.
- popularity : The popularity of the track. The value will be between 0 and 100, with 100 being the most popular
- id : Unique ID for track within ReccoBeats API
- href : Known spotify URLs for this track
- acousticness : Acousticness refers to how much of a song or piece of music is made up of natural, organic sounds rather than synthetic or electronic elements. In

other words, it's a measure of how "acoustic" a piece of music sounds. A confidence measure from 0.0 to 1.0, greater value represents higher confidence the track is acoustic.

- danceability : Danceability is a measure of how suitable a song is for dancing, ranging from 0 to 1. A score of 0 means the song is not danceable at all, while a score of 1 indicates it is highly danceable. This score takes into account factors like tempo, rhythm, beat consistency, and energy, with higher scores indicating stronger, more rhythmically engaging tracks.
- energy : Energy in music refers to the intensity and liveliness of a track, with a range from 0 to 1. A score of 0 indicates a very calm, relaxed, or low-energy song, while a score of 1 represents a high-energy, intense track. It's influenced by elements like tempo, loudness, and the overall drive or excitement in the music.
- instrumentalness : Predicts whether a track contains no vocals. "Ooh" and "aah" sounds are treated as instrumental in this context. Rap or spoken word tracks are clearly "vocal". The closer the instrumentalness value is to 1.0, the greater likelihood the track contains no vocal content. Values above 0.5 are intended to represent instrumental tracks, but confidence is higher as the value approaches 1.0.
- key : The key the track is in. Integers map to pitches using standard Pitch Class notation. E.g. 0 = C, 1 = C-sharp/D-flat, 2 = D, and so on. If no key was detected, the value is -1.
- liveness : Detects the presence of an audience in the recording. Higher liveness values represent an increased probability that the track was performed live. A value above 0.8 provides strong likelihood that the track is live.
- loudness : The overall loudness of a track in decibels (dB). Loudness values are averaged across the entire track and are useful for comparing relative loudness of tracks. Loudness is the quality of a sound that is the primary psychological correlate of physical strength (amplitude). Values typical range between -60 and 0 db.
- mode : Mode indicates the modality (major or minor) of a track. Major is represented by 1 and minor is 0.
- speechiness : Speechiness detects the presence of spoken words in a track. The more exclusively speech-like the recording (e.g. talk show, audio book, poetry), the closer to 1.0 the attribute value. Values above 0.66 describe tracks that are probably made entirely of spoken words. Values between 0.33 and 0.66 describe tracks that may contain both music and speech, either in sections or layered, including such cases as rap music. Values below 0.33 most likely represent music and other non-speech-like tracks.
- tempo : The overall estimated tempo of a track in beats per minute (BPM). Values typical range between 0 and 250
- valence : Valence in music measures the emotional tone or mood of a track, with a range from 0 to 1. A score of 0 indicates a song with a more negative, sad, or dark feeling, while a score of 1 represents a more positive, happy, or uplifting mood. Tracks with a high valence tend to feel joyful or energetic, while those with a low valence may evoke feelings of melancholy or sadness.

C. Method for Answering Research Questions

To answer our research questions, we combine descriptive statistics with inferential statistical tests and a predictive model. First, we evaluate whether metal main genres differ in their acoustic characteristics by applying one-way ANOVA separately for each continuous feature (acousticness, danceability, energy, instrumentalness, loudness, speechiness, tempo, and valence). For each ANOVA, we test the null hypothesis that all genre means are equal, and we report the corresponding p-value and effect size $(\eta 2)$ to quantify practical significance.

When ANOVA indicates a significant difference, we further investigate which genre pairs differ using post-hoc pairwise comparisons with a multiple-testing correction to control false positives. This supports comparisons between musically related genres (e.g., Black vs Death, Doom vs Post-metal) and helps interpret whether "similar" subgenres are acoustically close.

For the binary attribute mode (major=1, minor=0), a mean-comparison approach is not ideal. Instead, we treat mode as a categorical outcome and use a chi-square test of independence to test whether the proportion of major/minor tracks is independent of genre.

Finally, to study the relationship between acoustic traits and Spotify popularity, we fit a multiple linear regression (OLS) model where popularity is the dependent variable and the predictors include both acoustic features and genre (encoded as categorical variables). This allows us to estimate which features have statistically significant associations with popularity while controlling for genre effects.

III. Results

Descriptive Analysis

Within the data set we have the following data:
Number of Tracks: 2673
Number of Main Genres: 21
Number of Sub Genres: 52
Number of Unique Artists: 261

There are 21 main genres, and some of these genres also have subgenres, resulting in a total of 52 subgenres. Keep in mind that a main genre also has a subgenre with the same name. In other words, a main genre has at least one subgenre with the same name and at most n subgenres. A subgenre emerges when one genre merges with another or adopts a different artistic style.

This table will show how our data set was created, which main genres and subgenres are included, and how many songs belong to each subgenre. Some songs may belong to more than one subgenre, but we tried to keep this to a minimum in the data set.

Table 1
TABLE I
TRACK COUNT PER MAIN GENRE AND SUBGENRE

main genre	sub genre	track count
Alternative Metal	Alternative Metal	75
Alternative Metal	Funk Metal	67
Alternative Metal	Nu Metal	79
Alternative Metal	Rap Metal	39
Avantgarde Metal	Avantgarde Metal	59
Black Metal	1st Wave of Black Metal	40
Black Metal	2nd Wave of Black Metal	40
Black Metal	Blackgaze	19
Black Metal	Symphonic Black Metal	25
Black Metal	Viking Metal	60
Black Metal	War Metal	20
Christian Metal	Christian Metal	30
Death Metal	Blackened Death Metal	49
Death Metal	Death 'N' Roll	50
Death Metal	Death Metal	69
Death Metal	Melodic Death Metal	67
Death Metal	Symphonic Death Metal	49
Death Metal	Technical Death Metal	34
Doom Metal	Doom Metal	39
Doom Metal	Drone Metal	9
Doom Metal	Funeral Doom	28
Doom Metal	Sludge Metal	30
Doom Metal	Stoner Metal	20
Folk Metal	Celtic Metal	30
Folk Metal	Folk Metal	50
Folk Metal	Pagan Metal	30
Folk Metal	Pirate Metal	29
Glam Metal	Glam Metal	116
Gothic Metal	Gothic Metal	100
Grindcore	Grindcore	60
Hard Rock	Hard Rock	128
Heavy Metal	Heavy Metal	68
Heavy Metal	New Wave of British Heavy Metal	40
Heavy Metal	New Wave of Traditional Heavy Metal	29
Industrial Metal	Industrial Metal	54
Industrial Metal	Neue Deutsche Harte	48
Metalcore	Deathcore	65
Metalcore	Electronicore	14
Metalcore	Mathcore	16
Metalcore	Melodic Metalcore	75
Metalcore	Metalcore	133
Neoclassical Metal	Neoclassical Metal	29
Post-metal	Post-metal	18
Power Metal	Power Metal	95
Progressive Metal	Djent	28
Progressive Metal	Progressive Metal	98
Progressive Metal	Progressive Metalcore	28
Speed Metal	Speed Metal	20
Symphonic Metal	Symphonic Metal	59
Thrash Metal	Crossover Thrash	29
Thrash Metal	Groove Metal	76
Thrash Metal	Thrash Metal	111

A. *Popularity*

The boxplot in Figure 2 shows the distribution of track popularity for each main metal genre. The variability within genres is relatively high, indicating that popularity is influenced not only by genre characteristics but also by external factors such as release date, production quality, and the mainstream appeal of individual bands.

Genres like Alternative Metal, Hard Rock, and Thrash Metal display noticeably higher medians and wider upper quartiles, suggesting that these genres reach a broader audience and maintain more consistent mainstream visibility. In contrast, subgenres such as Christian Metal, Folk Metal, and Post-Metal show lower median popularity values and more compact distributions, reflecting their niche status and more limited listener base.

Overall, the popularity distributions demonstrate that metal genres differ widely in their accessibility and commercial reach, with some attracting a broader audience while others remain highly specialized.

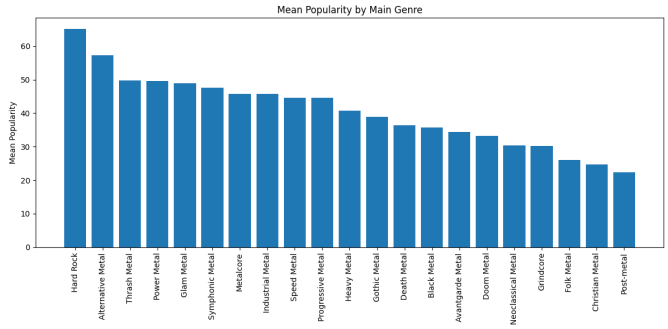

Fig. 1. Mean popularity comparison among metal main genres.

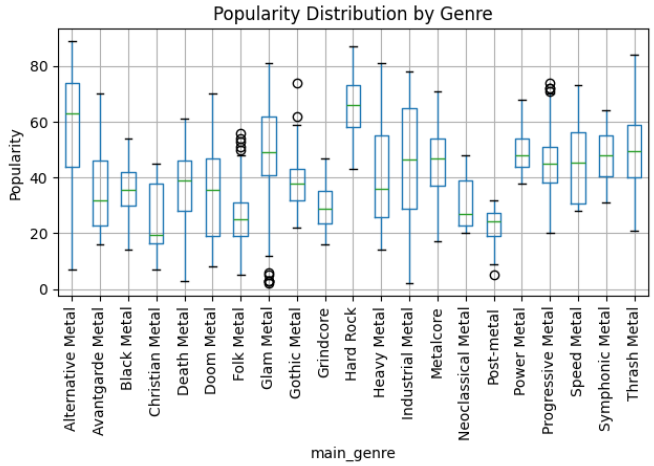

Fig. 2. Popularity distribution of tracks across metal main genres.

The bar chart illustrating the mean popularity values of metal subgenres reveals several important insights about how different stylistic branches of metal music perform on mainstream streaming platforms.

1. Most Popular Subgenres: Nu Metal, Hard Rock and Alternative Metal Lead by a Wide Margin The highest mean popularity values belong to Nu Metal, followed closely by Hard Rock and Alternative Metal. There are several obvious reasons for this:
Nu Metal (Korn, Slipknot, Linkin Park): It experienced a global explosion in the 2000s and is still listened to by a large

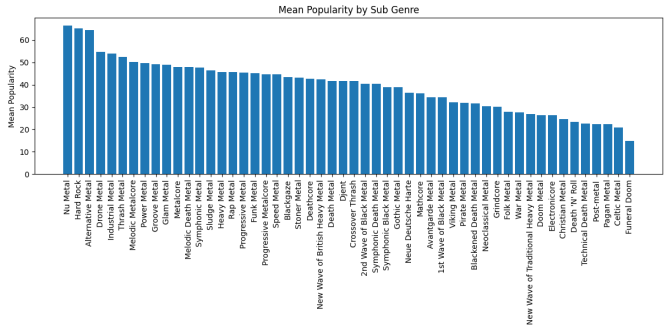

Fig. 3. Mean popularity comparison among metal sub genres.

audience today. Since a large portion of Spotify's listener base grew up during this period, its popularity remains high.
Hard Rock (AC/DC, Guns N' Roses, Aerosmith): Although it shares similar roots with heavy metal, hard rock is more rock than metal so it appeals to a much wider audience and its popularity is above all other metal subgenres.
Alternative Metal: Similar reasons as nu metal
This graph shows that subgenres with a more mainstream orientation have a clear advantage on Spotify.

2. Middle Segment: Thrash, Power Metal, Industrial Metal, Metalcore
The middle section of the graph features genres that are still popular but do not have as wide an audience reach as Nu Metal: Thrash Metal (thanks to Metallica's very high popularity)
Power Metal (especially with a large European-based fan base)
Industrial Metal (Rammstein's influence is evident)
Metalcore (especially popular among the modern Spotify audience)
These genres maintain balanced popularity levels due to their combination of broad appeal and dedicated fan bases.

3. Lower Ranks: Extreme Metal Genres (Black, Death, Doom, etc.) The lower section of the graph shows subgenres with relatively low popularity:
Black Metal (1st Wave, 2nd Wave, Blackgaze, Symphonic Black)
Death Metal (Technical, Melodic, Death 'N' Roll)
Doom / Funeral Doom
Folk, Pagan, Pirate Metal
Neoclassical Metal
Grindcore / Mathcore
The main reasons for the low popularity of these genres: More niche, extreme, or dark themes Less radio-friendly structure The majority of the Spotify audience does not actively listen to these genres and some are performed by musicians who consciously choose to remain underground It is expected that Funeral Doom has the lowest popularity, as the genre's tempo, atmosphere, and aesthetic approach appeal entirely to an "underground" audience.

B. Energy

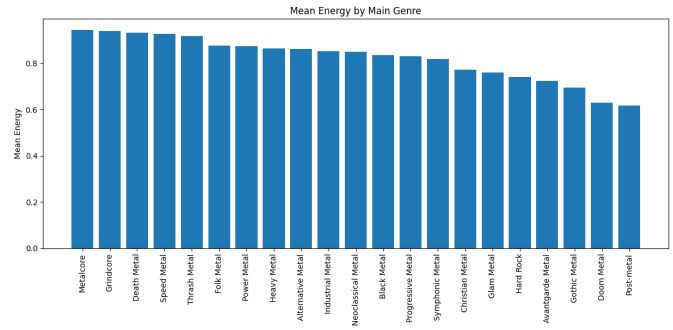

Fig. 4. Mean energy comparison among metal main genres.

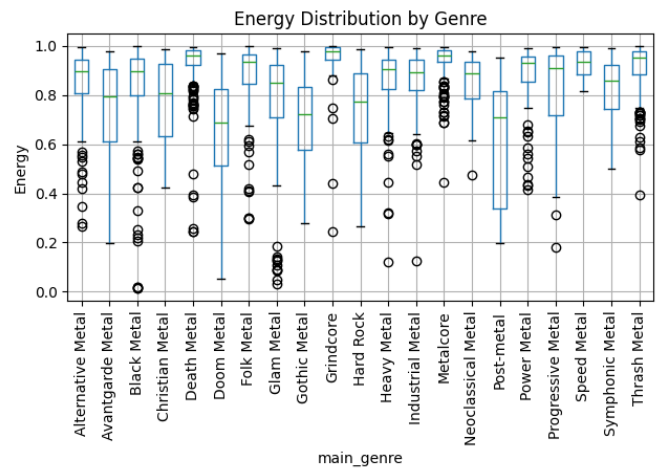

Fig. 5. Energy distribution among metal main genres.

The energy analysis reveals a clear separation between highly aggressive metal genres and more atmospheric or experimental styles. Metalcore, Grindcore, Death Metal, Speed Metal, and Thrash Metal occupy the top of the ranking, all exceeding a mean energy value above 0.90. These genres are structurally built on fast tempos, heavily distorted guitars, intense drumming, and aggressive vocal techniques, which aligns with their high energy scores.

Genres such as Power Metal, Folk Metal, Heavy Metal, and Industrial Metal follow slightly behind, showing moderately high energy values (0.85–0.88). Their sound is still dynamic and loud, but often incorporates more melodic or rhythmic variation.

Toward the lower end of the spectrum, we observe genres that traditionally rely on atmosphere or emotional expression rather than aggression. Avantgarde Metal, Gothic Metal, Doom Metal, and especially Post-metal show the lowest mean energy values. Post-metal in particular is characterized by long, slow-building structures and ambient textures, which explains its significantly lower energy score.

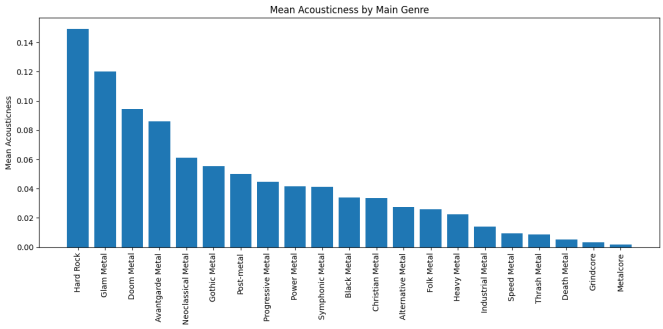

Fig. 6. Mean acousticness comparison among metal main genres.

C. Acousticness

The graph clearly shows that metal genres generally have very low acousticness values. The main reason for this is that metal music is based on highly distorted guitars, powerful amplification, and intense electronic production.We did not use a boxplot for this value because most values are already extremely low.

However, genres such as Hard Rock, Glam Metal, and to some extent Doom Metal have higher average acousticness values than other metal genres. These genres sometimes feature acoustic guitar intros, clean-toned sections, or more traditional rock influences. Similarly, Avant-Garde and Neoclassical Metal have slightly higher average acousticness values due to their structural traces of classical music or experimental elements.

On the other hand, genres such as Metalcore, Grindcore, Death Metal, and Thrash Metal, which are more intense, aggressive, and distortion-heavy, have acousticness values close to zero. This stems from these genres relying almost entirely on electronically processed and high-energy productions.

Overall, the graph shows that acousticness is a distinguishing but limited feature among metal genres; it only creates meaningful differences in more melodic or traditional rock-based genres.

D. Instrumentalness

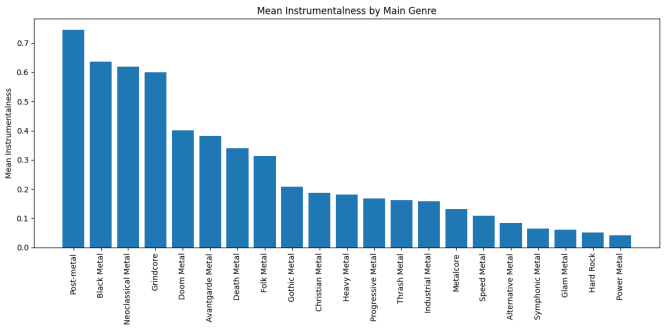

Fig. 7. Mean instrumentalness comparison among metal main genres.

The instrumentalness feature measures the likelihood that a track contains no vocals, with values close to 1.0 indicating fully instrumental compositions. The results reveal a highly uneven distribution of instrumentalness across metal genres, reflecting the stylistic and structural diversity within metal music. Post-metal stands out with the highest mean instrumentalness, exceeding 0.70. This aligns well with the genre's aesthetic: long atmospheric passages, minimal vocals, and an emphasis on texture and dynamic buildup. Similarly, Black Metal, Neoclassical Metal, and Grindcore show relatively high instrumentalness values. In Black Metal, this is likely due to raw production styles where vocals blend into the mix and may be detected as "non-vocal noise" by the audio model. Neoclassical Metal often features extended instrumental virtuosity inspired by classical music. Grindcore, despite being extremely aggressive, frequently includes chaotic sections where vocals are brief, unintelligible, or absent. In contrast, genres such as Power Metal, Hard Rock, Glam Metal, and Symphonic Metal exhibit very low instrumentalness, typically below 0.10. These genres rely heavily on clean, prominent vocals, melodic choruses, and narrative lyrics—elements that sharply reduce instrumentalness scores. Genres in the middle range, including Death Metal, Doom Metal, and Gothic Metal, reflect mixed vocal-instrumental characteristics. For example, harsh growls or guttural vocals in Death Metal might be partially misclassified as instrumental sections.

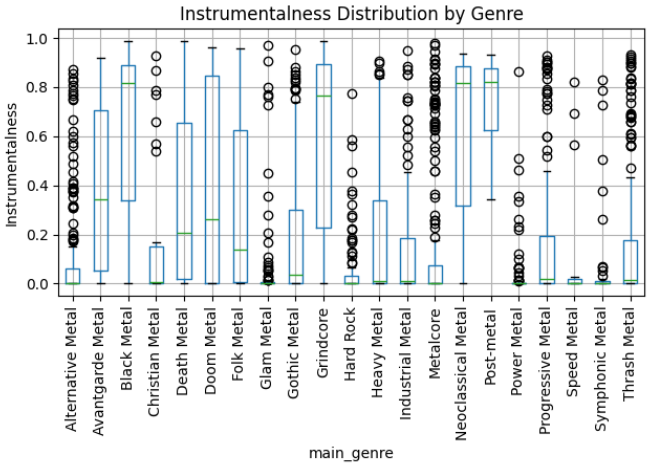

Fig. 8. Instrumentalness distribution among metal main genres.

Overall Conclusion Instrumentalness is one of the clearest acoustic indicators separating different metal genres. Genres with atmospheric, experimental, or classical influences (e.g., Post-metal, Neoclassical Metal) tend to exhibit high instrumentalness. Genres centered on melody, vocals, and structured songwriting (e.g., Power Metal, Glam Metal) show consistently low values. This distribution highlights how vocal presence—or the lack of it—is a defining characteristic that strongly differentiates metal subgenres.

E. Danceability

Industrial Metal is the most danceable genre, likely because it uses electronic elements and steady rhythmic patterns. Alternative Metal and Hard Rock also show relatively high danceability due to their more accessible grooves.
On the other hand, extreme genres like Black Metal, Grindcore, and Doom Metal have the lowest danceability, which fits their chaotic structures, harsh textures, and irregular rhythms. The boxplot shows that genres such as Industrial and Alternative Metal have wide variation, meaning some songs are much more danceable than others, while extreme genres remain consistently low.

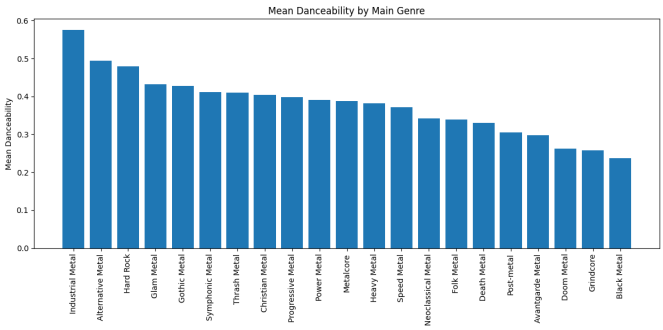

Fig. 9. Mean danceabilitys comparison among metal main genres.

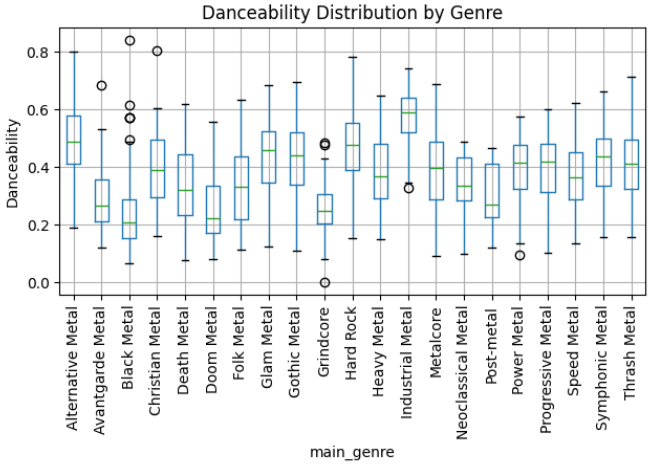

Fig. 10. Danceability distribution among metal main genres.

F. Valence

The valence distribution reveals a strong emotional divide across metal genres. Hard Rock stands out with the highest mean valence, indicating a more upbeat, positive, and energetic emotional tone. This aligns well with its traditional emphasis on catchy riffs, major-key progressions, and overall accessible sound.

Genres like Alternative Metal, Industrial Metal, Glam Metal, and Christian Metal also show relatively high valence values,

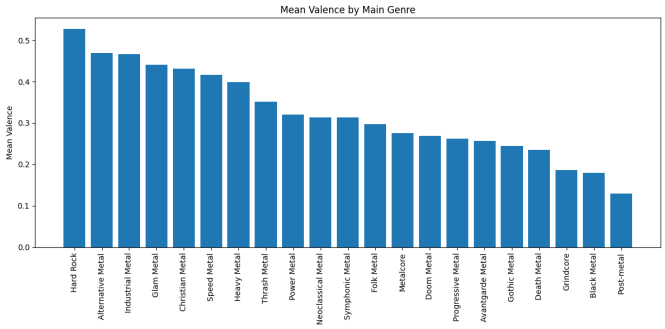

Fig. 11. Mean valence comparison among metal main genres.

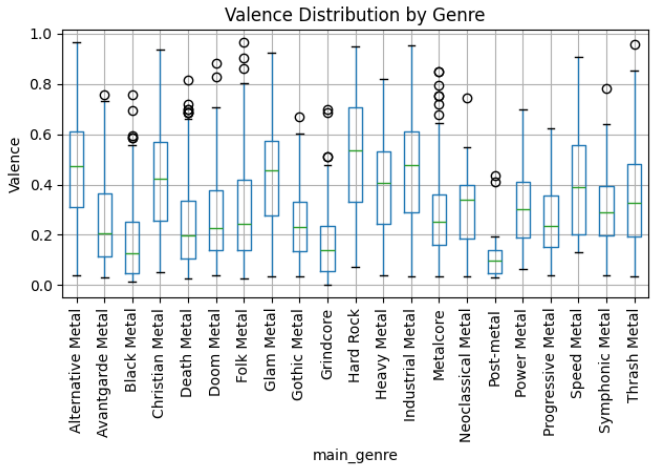

Fig. 12. Valence distribution among metal main genres.

suggesting they incorporate brighter harmonies, cleaner production, or more melodic structures compared to the rest of the metal spectrum.

In contrast, genres such as Post-metal, Black Metal, Grindcore, and Death Metal exhibit the lowest mean valence. These styles tend to favor darker tonalities, dissonance, slower tempos, or aggressive atmospheres—musical traits the Spotify model interprets as "negative" or "sad" emotional content.

Overall, the valence analysis supports the intuitive emotional spectrum of metal:
Mainstream-leaning or melodic genres → higher valence
Extreme or atmospheric genres → lower valence
This reinforces how emotional expression in metal varies dramatically between subcultures, from uplifting rock-influenced styles to the bleak and intense aesthetics of extreme metal.

G. Loudness

The loudness results show clear production differences across metal genres. Modern and highly produced styles—such as Metalcore, Speed Metal, Symphonic Metal, Alternative Metal, and Power Metal—are the loudest in the dataset, typi-

cally sitting around –4 to –6 dB, reflecting heavy compression and contemporary mastering trends.

Mid-range genres like Industrial Metal, Thrash Metal, Heavy Metal, and Progressive Metal fall near –6 to –8 dB, balancing power with some dynamic range.

The quietest genres are Doom Metal, Post-metal, Avant-garde Metal, Neoclassical Metal, and Christian Metal, which often reach –9 dB or lower, matching their more atmospheric, raw, or experimental sound profiles.

Overall, the loudness levels align closely with each genre's stylistic identity: polished modern subgenres are significantly louder, while atmospheric or old-school styles retain greater dynamic variation.

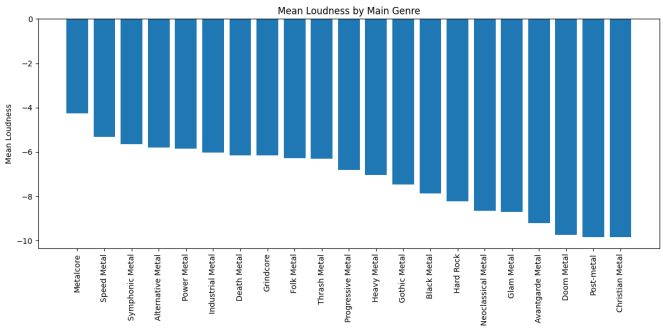

Fig. 13. Mean loudness comparison among metal main genres.

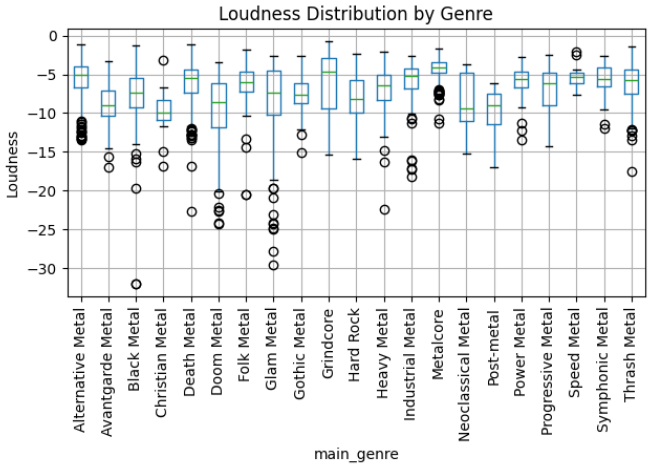

Fig. 14. Loudness distribution among metal main genres.

H. Speechiness

Across all metal genres, speechiness values remain generally low, which is expected since metal music is dominated by sung or screamed vocals rather than spoken-word passages.

A few genres stand out:

Grindcore shows the highest speechiness, reflecting its chaotic vocal style, use of spoken samples, and fast, abrasive delivery.

Metalcore and Death Metal also rank high, likely due to breakdown callouts, spoken intros, or aggressive rhythmic vocal phrasing.

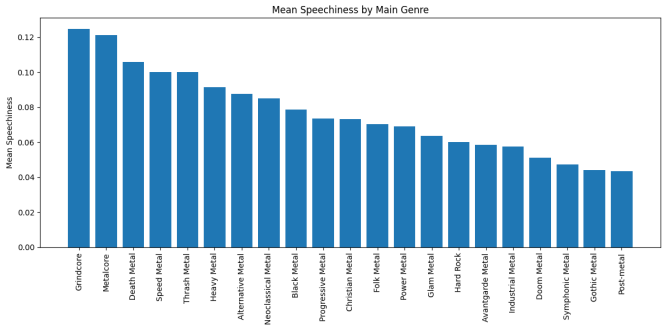

Fig. 15. Mean speechiness among metal main genres.

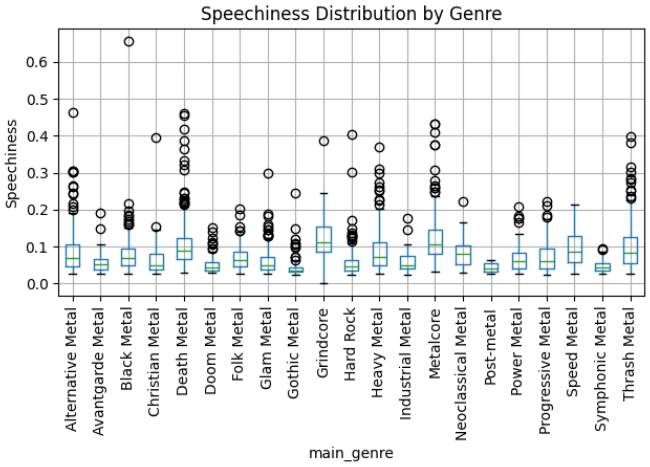

Fig. 16. Speechiness distribution among metal main genres.

Mid-range genres such as Thrash, Speed Metal, and Heavy Metal show moderate speechiness but still remain low overall, reflecting mostly melodic or shouted vocals.

Genres with the lowest speechiness include Gothic Metal, Symphonic Metal, Industrial Metal, Doom Metal, and Post-metal, which rely heavily on atmospheric, melodic, or instrumental sections rather than speech-like vocals.

Overall conclusion: Speechiness is generally low across metal, confirming that spoken-word elements are rare. Only a few extreme genres (mainly Grindcore and Metalcore) incorporate notable speech-like or sample-based components.

I. Tempo

Across metal genres, average tempo values cluster tightly, typically between 118–135 BPM, indicating a relatively stable tempo range across the metal spectrum. Still, some notable distinctions appear:

Power Metal and Neoclassical Metal show the highest mean tempos, around 133–135 BPM, reflecting their emphasis on fast, energetic, and technically demanding compositions.

Metalcore, Heavy Metal, Black Metal, Thrash Metal, and Speed Metal also maintain high BPM levels, consistent with

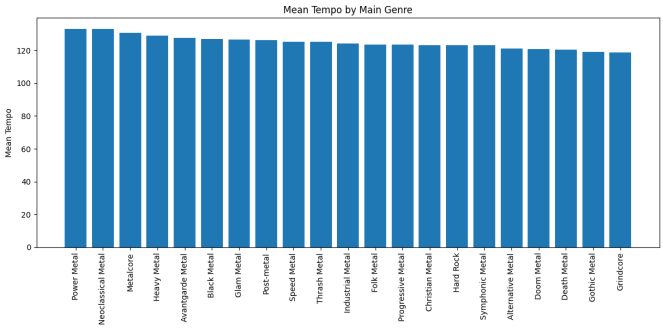

Fig. 17. Mean tempo among metal main genres.

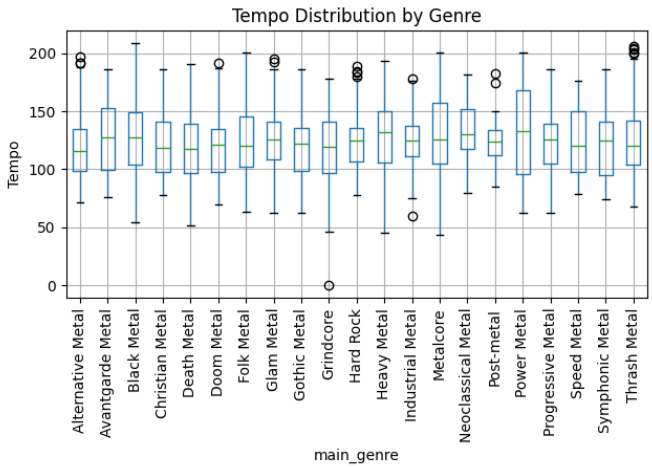

Fig. 18. Tempo distribution among metal main genres.

their aggressive rhythmic drive.
Genres like Gothic Metal, Doom Metal, Death Metal, and Grindcore fall toward the lower end (around 118–122 BPM), fitting their darker, heavier, or more atmospheric characteristics.

The boxplots reveal significant variability within each genre, meaning tempo is not a strict defining factor for most metal subgenres—but overall, fast-paced rhythmic intensity remains a core trait of metal music.

J. Mode

Major–Minor Proportion Plot – Interpretation

This plot shows how each metal subgenre is distributed in terms of track mode (major vs minor). Overall: Subgenres dominated by Minor (darker / more melancholic tone) These genres have more than 50 percent minor tracks:

Gothic Metal (62)
Power Metal (60)
Symphonic Metal (58)
Folk Metal (57)
Avantgarde Metal (56)

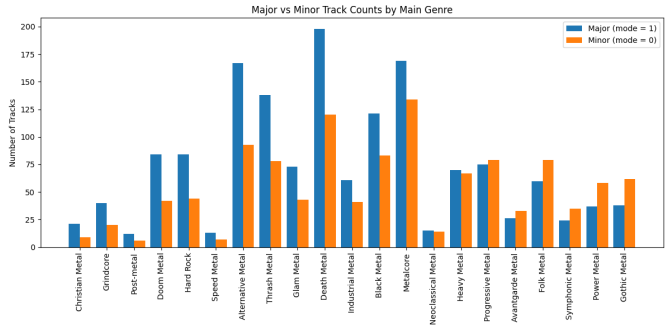

Fig. 19. Major and minord accord count for each genre.

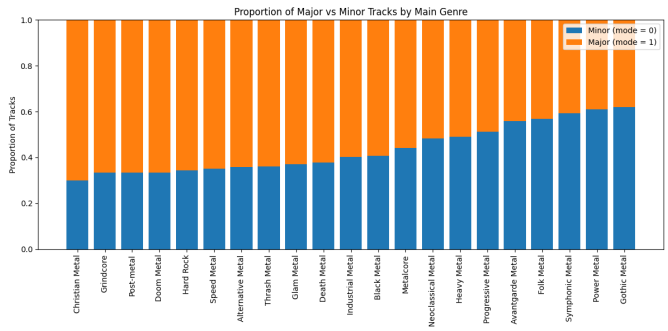

Fig. 20. Ratios of major to minor accords for each genre.

Progressive Metal (56)
Heavy Metal (50+)

This aligns well with their musical character — dramatic, atmospheric, and often emotionally heavy.

Subgenres dominated by Major (brighter / more energetic tone) Genres with 60 percent + major tracks:
Christian Metal (the most major-heavy genre)
Grindcore
Post-metal
Doom Metal
Hard Rock
Speed Metal
Alternative Metal

It's especially interesting that Grindcore, despite its extreme aggression, favors major mode tonally.

Analysis for Answering Reseach Questions

Overview of the analytical strategy

To answer our main research question—whether metal genres are acoustically distinct in a meaningful way—we conducted inferential analyses on the audio-feature variables of our curated dataset. Since descriptive statistics alone cannot establish whether observed differences are statistically reliable, we complemented them with:

(i) One-way ANOVA to test whether the mean of each continuous feature differs across main genres,

(ii) post-hoc pairwise comparisons to identify which genre pairs drive the differences,

(iii) a Chi-square test of independence for the binary variable mode (major vs minor), and

(iv) multiple linear regression to study the relationship between popularity and the audio features.

All tests were conducted at a significance level of $\alpha = 0.05$. When multiple genre-pair comparisons were needed, we relied on post-hoc procedures to control for inflated Type-I error across many comparisons.

Main Research Question — Are metal genres acoustically different in a meaningful way?

We treated each audio feature (e.g., energy, danceability, acousticness, instrumentalness, loudness, speechiness, tempo, valence) as a continuous outcome and applied a one-way ANOVA with main genre as the categorical factor:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \quad \text{vs.} \quad H_1 : \exists\, i \neq j \text{ such that } \mu_i \neq \mu_j$$

Across the majority of examined audio features, one-way ANOVA results demonstrate that main genre membership explains a statistically significant portion of variance ($p < 0.001$), indicating that metal genres are not only descriptively distinct but also separable in a statistically meaningful manner. Effect size analysis further reveals that this genre-related signal is particularly strong for features such as danceability, acousticness, instrumentalness, loudness, valence, speechiness, and energy, while tempo and liveness exhibit comparatively weaker—yet still significant—differences across genres. Collectively, these findings provide robust empirical evidence that metal genres possess measurable acoustic distinctions, contradicting the common claim that "all metal sounds the same."

Sub-Question — Which features differ the most across genres, and between which genre pairs?

A statistically significant ANOVA indicates that at least one group differs, but does not specify where the differences occur. Therefore, for each feature with a significant ANOVA result, we performed post-hoc pairwise comparisons between genres. This allows us to identify:

- which genre pairs have significant mean differences
- which features are most discriminative across the genre space.

Post-hoc comparisons further illustrate concrete separations between specific genre pairs. For example, Death Metal and Black Metal differ significantly in multiple dimensions (including instrumentalness, energy, loudness, and speechiness), suggesting that historically adjacent "extreme" genres can still occupy clearly distinct acoustic profiles. Similarly, Doom Metal and Post-metal exhibit significant differences in instrumentalness and valence, aligning with their contrasting stylistic emphasis.

TABLE II
ONE-WAY ANOVA RESULTS BY FEATURE (FEATURE MAIN GENRE).

feature	n	k_genres	F	p_value	eta_squared
danceability	2673	21	61.028	1.058e-200	0.315
instrumentalness	2673	21	53.119	4.578e-177	0.286
energy	2673	21	43.853	3.860e-148	0.249
valence	2672	21	42.064	2.195e-142	0.241
loudness	2673	21	31.303	1.126e-106	0.191
speechiness	2673	21	26.056	1.580e-88	0.164
acousticness	2673	21	13.942	6.583e-45	0.095
liveness	2673	21	2.844	2.471e-05	0.021
tempo	2673	21	2.411	4.316e-04	0.018

Sub-Question — Are "similar" subgenres actually close acoustically?

To evaluate whether historically or perceptually similar styles are also close in feature space, we inspected the post-hoc results for representative "neighbor" comparisons (e.g., Death vs Black, Doom vs Sludge, etc.). The results suggest that similarity depends on the feature: two genres may appear close on one dimension (e.g., tempo) while diverging on another (e.g., valence or loudness). This supports the idea that genre similarity in metal is multi-dimensional rather than reducible to a single attribute.

Table V reports selected pairwise comparisons between metal genres using Welch's t-test with Bonferroni correction. The results indicate that several genre pairs exhibit statistically significant differences across multiple acoustic features even after strict multiple-comparison correction, confirming that the observed inter-genre differences are robust.

Notably, comparisons between Death Metal and Black Metal show consistent and significant differences in instrumentalness, danceability, energy, loudness, speechiness, valence, and tempo, with moderate to large effect sizes (Cohen's d). This suggests that despite historical and aesthetic similarities, these two genres differ substantially in measurable acoustic properties.

Similarly, Doom Metal vs. Post-metal comparisons reveal strong contrasts, particularly in instrumentalness and valence, reflecting Post-metal's more atmospheric and instrumental-focused structure. Differences observed between Industrial Metal and Alternative Metal, especially in danceability and speechiness, highlight the influence of electronic and rhythmic elements in Industrial Metal.

Overall, the pairwise analysis reinforces the ANOVA findings by demonstrating that genre-level differences are not only global but also persist at the level of specific genre pairs and individual audio features.

Sub-Question — Is musical mode (major/minor) associated with genre?

Unlike other audio features, mode is binary (major = 1, minor = 0), which makes mean-comparison ANOVA less appropriate as a primary tool. Instead, we tested whether mode distribution depends on genre via a Chi-square test of independence:

H_0 : mode is independent of genre vs. H_1 : depends on gen

A chi-square test of independence was conducted to examine the relationship between harmonic mode (major/minor) and metal genre. The results indicate a statistically significant association between mode and genre ($\chi^2 = 85.49$, $df = 20$, $p < 0.001$, $n = 2673$).

TABLE III
CHI-SQUARE TEST RESULTS FOR MODE AND GENRE

Statistic	Value
χ^2 statistic	85.49
Degrees of freedom (df)	20
p-value	< 0.001
Sample size (n)	2673

The Chi-square results indicate that the proportion of major vs minor tracks is not randomly distributed across genres; some genres show a stronger tendency toward minor-mode compositions, while others contain relatively more major-mode tracks. This finding is consistent with genre-specific aesthetics (e.g., darker harmonic language in certain extreme styles versus brighter tonal choices in more melodic or classic-oriented categories).

Sub-Question — Is popularity related to acoustic characteristics?

To test whether audio features predict Spotify popularity, we fit a multiple linear regression model with popularity as the dependent variable and the audio features as predictors:

$$\text{popularity} = \beta_0 + \sum_m \beta_m x_m + \varepsilon$$

We evaluated model performance using standard regression metrics (e.g.,R2, MAE, RMSE) and interpreted the learned coefficients to determine which features have positive or negative associations with popularity. The results suggest that popularity can be partially explained by acoustic features, but only to a limited extent (as expected, since popularity also depends on external factors such as marketing, era, platform dynamics, and artist-level effects).

The Actual vs. Predicted Popularity scatter plot shows a clear positive relationship between observed and predicted values, indicating that the linear regression model captures a meaningful portion of the variability in popularity. However, the noticeable spread around the diagonal line reveals substantial unexplained variance, suggesting that while acoustic features contribute to popularity, they are not sufficient to fully predict it. This behavior is consistent with the expectation that popularity is also influenced by non-audio factors such as cultural context, artist reputation, and external promotion.

The regression coefficient table provides further insight into which variables drive these predictions. Loudness exhibits a statistically significant positive coefficient, indicating that louder tracks tend to be associated with higher popularity scores. In contrast, many genre dummy variables (e.g., Death Metal, Black Metal, Doom Metal, Grindcore) have large and significant negative coefficients relative to the reference category, implying systematically lower popularity levels for these genres when controlling for acoustic features. This highlights that genre membership itself plays a strong role beyond low-level audio characteristics.

Overall, the regression analysis supports the conclusion that acoustic properties have a measurable but limited explanatory power for popularity. The model is statistically sound and interpretable, yet its moderate predictive performance reinforces the idea that popularity is a multifactorial phenomenon extending beyond purely musical attributes.

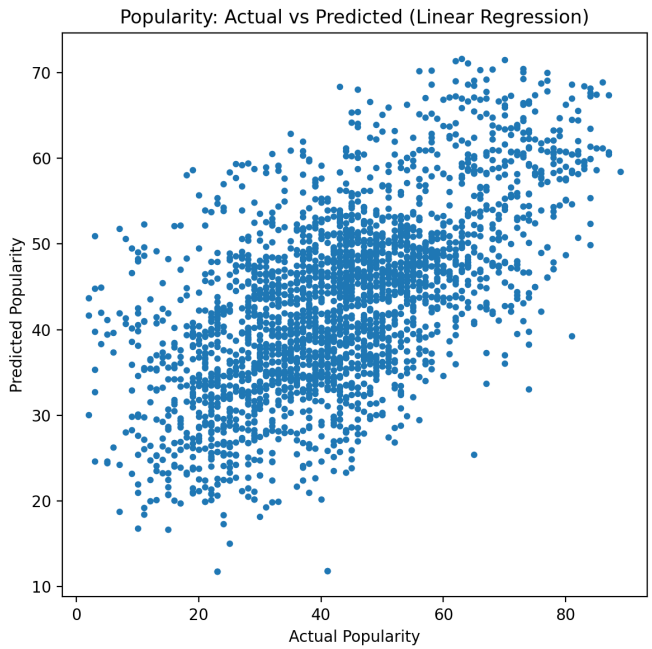

TABLE IV
MOST SIGNIFICANT REGRESSION COEFFICIENTS FOR PREDICTING POPULARITY.

term	beta	std_err	p_value
Intercept	70.4356	3.0955	8.216e-105
C(main_genre)[T.Folk Metal]	-29.9563	1.4549	1.744e-87
C(main_genre)[T.Death Metal]	-18.7142	1.2079	6.851e-52
C(main_genre)[T.Doom Metal]	-19.7215	1.5915	2.561e-34
C(main_genre)[T.Black Metal]	-17.2247	1.4400	3.771e-32
C(main_genre)[T.Grindcore]	-23.4397	2.0203	2.168e-30
C(main_genre)[T.Gothic Metal]	-17.7438	1.6267	3.979e-27
C(main_genre)[T.Christian Metal]	-27.8952	2.5833	1.253e-26
loudness	1.1985	0.1134	1.308e-25
C(main_genre)[T.Metalcore]	-11.8726	1.1745	1.353e-23
C(main_genre)[T.Heavy Metal]	-14.3372	1.4209	1.618e-23
C(main_genre)[T.Avantgarde Metal]	-18.7074	1.9900	1.133e-20
C(main_genre)[T.Post-metal]	-29.8988	3.3382	6.170e-19
C(main_genre)[T.Progressive Metal]	-11.5330	1.3904	1.711e-16
C(main_genre)[T.Neoclassical Metal]	-21.5343	2.6572	8.037e-16
C(main_genre)[T.Industrial Metal]	-12.0125	1.5676	2.529e-14
C(main_genre)[T.Hard Rock]	9.6091	1.4636	6.220e-11
C(main_genre)[T.Symphonic Metal]	-10.7901	1.9424	3.052e-08
C(main_genre)[T.Power Metal]	-7.7759	1.6241	1.780e-06
C(main_genre)[T.Thrash Metal]	-5.8679	1.2479	2.706e-06

TABLE V
SELECTED PAIRWISE COMPARISONS (WELCH T-TEST) WITH BONFERRONI CORRECTION.

genre_1	genre_2	feature	n1	n2	mean_1	mean_2	t_stat	p_value	p_bonf	cohens_d
Death Metal	Black Metal	instrumentalness	318	204	0.340	0.636	-9.661	4.651e-20	1.674e-18	-0.876
Death Metal	Black Metal	danceability	318	204	0.331	0.237	8.453	3.953e-16	1.423e-14	0.748
Industrial Metal	Alternative Metal	danceability	102	260	0.576	0.494	7.073	1.358e-11	4.890e-10	0.708
Death Metal	Black Metal	energy	318	204	0.933	0.834	6.815	5.964e-11	2.147e-09	0.693
Industrial Metal	Alternative Metal	speechiness	102	260	0.057	0.087	-6.582	1.651e-10	5.943e-09	-0.561
Death Metal	Black Metal	loudness	318	204	-6.153	-7.859	5.701	2.642e-08	9.513e-07	0.551
Doom Metal	Post-metal	instrumentalness	126	18	0.402	0.746	-6.496	3.467e-08	1.248e-06	-0.914
Death Metal	Black Metal	speechiness	318	204	0.106	0.079	5.187	3.152e-07	1.135e-05	0.448
Doom Metal	Post-metal	valence	125	18	0.269	0.130	4.383	1.356e-04	4.882e-03	0.817
Death Metal	Black Metal	valence	318	204	0.235	0.180	3.792	1.701e-04	6.123e-03	0.337
Thrash Metal	Speed Metal	loudness	216	20	-6.314	-5.329	-2.862	7.122e-03	2.564e-01	-0.393
Industrial Metal	Alternative Metal	instrumentalness	102	260	0.158	0.084	2.570	1.119e-02	4.030e-01	0.344
Death Metal	Black Metal	acousticness	318	204	0.005	0.034	-2.485	1.366e-02	4.916e-01	-0.267
Death Metal	Black Metal	tempo	318	204	120.607	126.965	-2.360	1.872e-02	6.740e-01	-0.215
Doom Metal	Post-metal	danceability	126	18	0.262	0.305	-1.556	1.337e-01	1.000e+00	-0.382

IV. CONCLUSION

Summary

This study set out to investigate whether metal music genres are acoustically distinct in a statistically meaningful way and to examine how these acoustic properties relate to popularity. Using a manually curated and balanced dataset of 2,673 metal tracks, enriched with Spotify popularity scores and detailed audio features, we applied a combination of descriptive statistics, inferential hypothesis testing, and regression modeling within the scope of introductory data analysis methodologies.

The descriptive analysis revealed substantial variation across metal genres in nearly all examined acoustic dimensions, including energy, danceability, instrumentalness, loudness, valence, speechiness, tempo, and mode. Visual tools such as boxplots, histograms, and mean comparison charts demonstrated that while some features (e.g., tempo) cluster within a relatively narrow range, others (e.g., energy, instrumentalness, and valence) provide clear separation between stylistically different genres. These findings already suggested that metal genres exhibit systematic acoustic patterns rather than random variation.

Inferential analyses strongly reinforced these observations. One-way ANOVA results showed that genre membership explains a statistically significant portion of variance across all major audio features, with particularly strong effects observed for danceability, instrumentalness, energy, loudness, valence, and speechiness. Post-hoc pairwise comparisons further demonstrated that even historically or perceptually similar genres—such as Death Metal and Black Metal, or Doom Metal and Post-metal—can differ significantly across multiple acoustic dimensions. Together, these results provide robust empirical evidence against the common claim that "all metal sounds the same," showing instead that metal genres are both descriptively and statistically distinguishable.

For the binary harmonic attribute mode (major vs. minor), a chi-square test of independence revealed that tonal modality is not uniformly distributed across genres. Certain genres display a clear preference for minor modes, aligning with darker or more melancholic aesthetics, while others show a higher proportion of major-mode compositions. This confirms that harmonic structure is another dimension through which genre identity manifests in metal music.

Finally, the multiple linear regression analysis explored the relationship between acoustic features and Spotify popularity. The results indicate that acoustic characteristics have a statistically significant but limited explanatory power for popularity. Features such as loudness show a positive association with popularity, while many genre indicators retain strong effects even after controlling for audio features. The actual versus predicted popularity plot highlights that, although the model captures meaningful trends, a substantial portion of popularity variance remains unexplained. This outcome is expected, as popularity is influenced by numerous external factors beyond musical structure, including cultural context, marketing, historical timing, and artist-level reputation.

Overall, this project demonstrates how foundational data analysis techniques—descriptive statistics, hypothesis testing, ANOVA, chi-square analysis, and linear regression—can be coherently combined to answer real-world research questions using complex cultural data. The findings confirm that metal genres possess distinct and measurable acoustic identities and that these identities interact with, but do not fully determine, mainstream popularity.

Future Work

Several directions for future research naturally emerge from this study. First, incorporating temporal information such as release year could help disentangle genre effects from historical production trends and changes in listener behavior. Second, extending the regression framework to non-linear models or machine learning approaches may improve predictive performance and capture more complex interactions between features. Third, incorporating lyrical content, timbral descriptors, or listener-level data could provide a richer, multimodal understanding of genre identity and popularity. Finally, expanding the dataset to include non-metal genres would allow for cross-genre comparisons and position metal within the broader musical landscape.